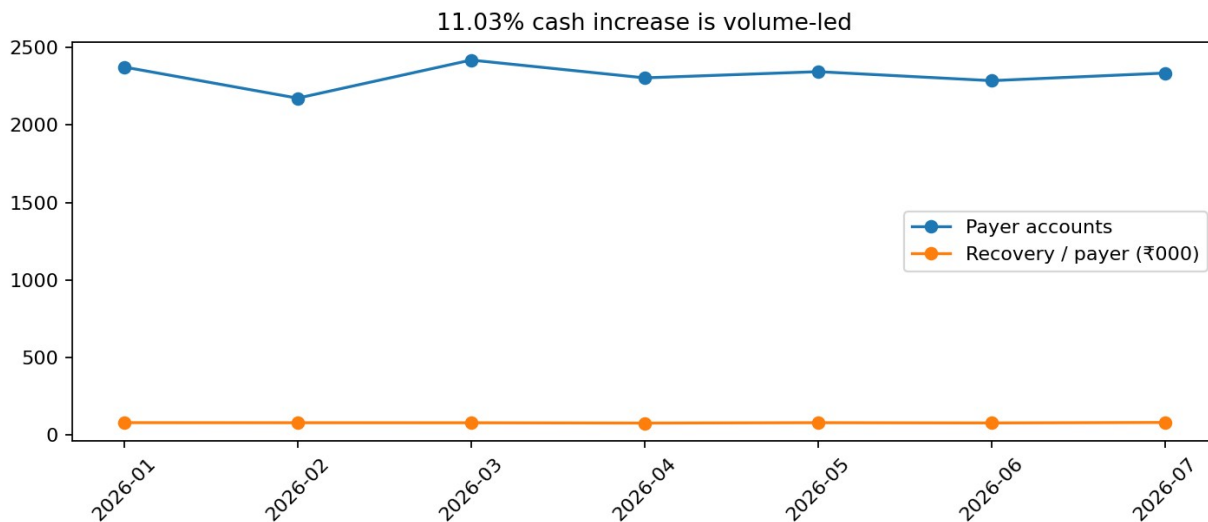


Collections Recovery Audit

Independent reconstruction, data forensics, driver analysis, statistical investigation, counterfactual methodology and ₹10 Cr recommendation.

Executive conclusion

The reported 11% month-on-month improvement is true only as the Feb→Mar increase in cleaned successful-payment cash (+11.03%). It is not evidence of an 11% improvement in collection effectiveness: payer accounts increased 11.32% while recovery per payer decreased 0.26%.



1. What happened?

month	recovery_cr	payers	recovery_per_payer_k	mom_cash_pct
2026-01	18.722912759	2374	78.86652383740523	nan
2026-02	17.014245375999998	2173	78.29841406350667	-9.126076722109676
2026-03	18.891237402	2419	78.09523522943365	11.031885249801654
2026-04	17.513804341	2304	76.01477578559027	-7.291386115629317
2026-05	18.42502785	2344	78.60506761945393	5.202887341083384
2026-06	17.555972697	2286	76.79778082677164	-4.716710118839796
2026-07	18.724226508	2335	80.18940688650964	6.654452197910032

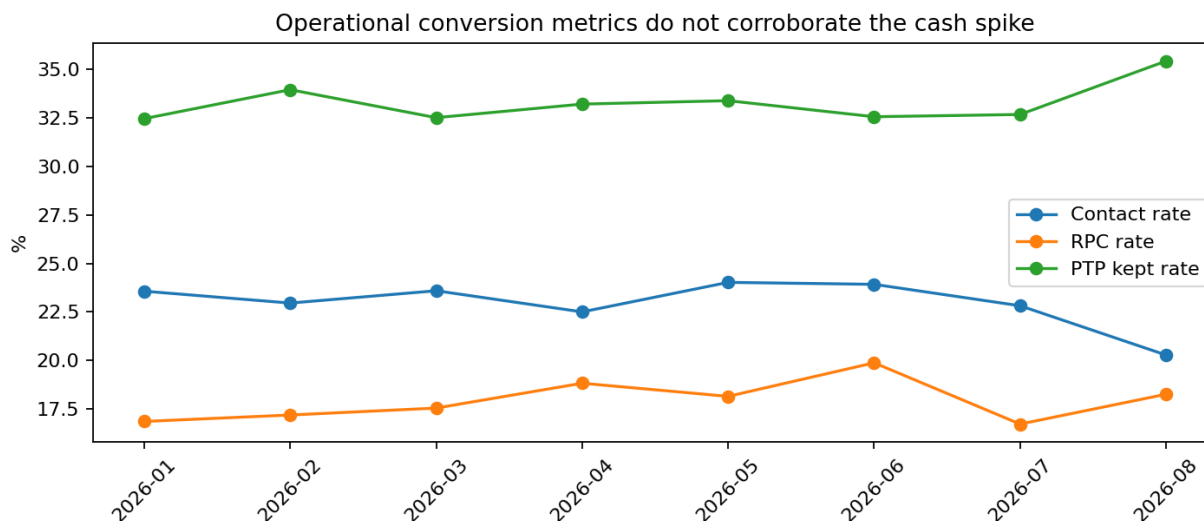
The supplied source period is 1 Jan 2026 to 8 Aug 2026; August is partial. A complete 12-month historical reconstruction is therefore not supported by the supplied extract.

2. Definitions

- Recovery = deduplicated payment_id SUCCESS value by event month.
- Recovery / payer = recovery / distinct payer accounts.
- Contact rate = distinct ANSWERED accounts / distinct attempted accounts.
- RPC rate = RPC-like disposition on ANSWERED calls / answered accounts.
- PTP kept = KEPT / (KEPT + BROKEN + CANCELLED).
- Recovery / agent-hour = clean recovery / collection-session hours.
- Cost / ₹ recovered is not computable because no cost ledger is supplied.

- Historical recovery rate vs opening outstanding is not reliable because periodized historical balances are absent.
- Channel conversion = descriptive 14-day post-exposure conversion, not causal ROI.

3. Operational evidence



Operational conversion metrics do not corroborate a broad effectiveness step-change. This is strong evidence against treating the +11.03% cash movement as an 11% operating improvement.

4. Data forensics

issue	business_impact
Duplicate payment IDs	500 duplicate IDs; ₹2.59 Cr duplicate SUCCESS value removed
Payment-reference collision	3407 references span multiple accounts
Account-borrower conflict: calls	91,342/91,350 (99.99%)
Account-borrower conflict: call_attempts	119,989/120,000 (99.99%)
Account-borrower conflict: call_dispositions	34,998/35,000 (99.99%)
Account-borrower conflict: payments	25,499/25,500 (100.00%)
Account-borrower conflict: PTP	17,999/18,000 (99.99%)
Account-borrower conflict: field_visits	24,998/25,000 (99.99%)
Account-borrower conflict: WhatsApp	60,596/60,600 (99.99%)
Account-borrower conflict: SMS	44,995/45,000 (99.99%)
Agent ID instability	1,000/1,000 agent IDs have conflicting identity fields
Duplicate call IDs	1,350 call IDs duplicated; 11 have conflicting timestamps

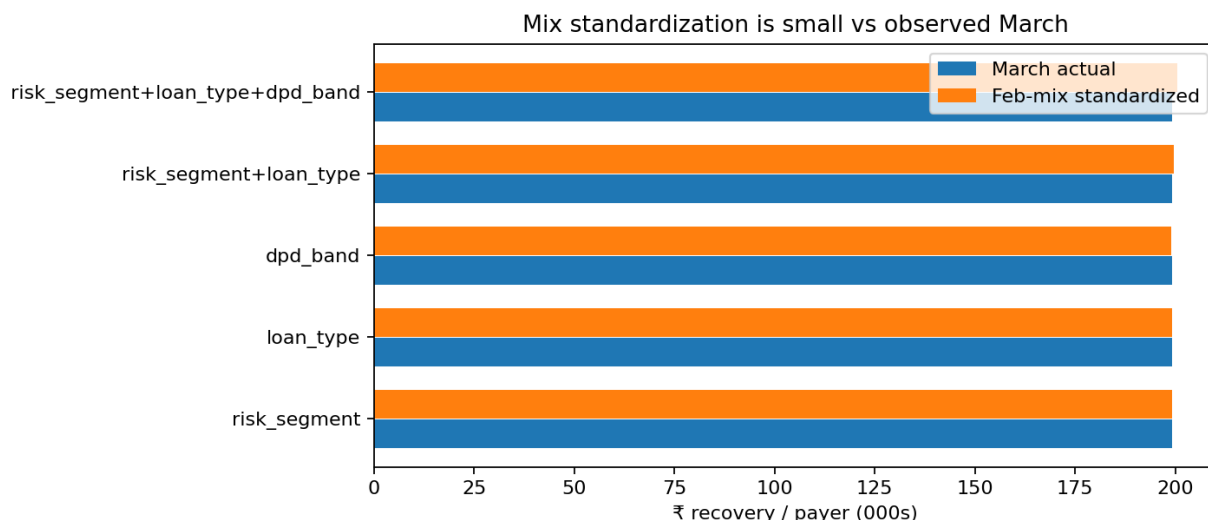
160 repeated account-date targeting keys are retained as multiple campaign exposures, not misclassified as duplicate data.

5. Drivers

Observable dimensions analyzed: loan type, risk, DPD, geography, borrower segment, campaign/channel, telephony vendor, calling time, attempt frequency and agent tenure. Client and language are absent from the supplied schemas and are not inferred. Historical agent attribution is low-confidence because agent_id is overwritten across identities.

state	recovery	payers	recovery_per_payer
Maharashtra	674082308.79	6169	109269.2995282866
Delhi	342399929.66	3677	93119.37167799838

Karnataka	334861446.51	3603	92939.61879267276
Haryana	332144774.43	3578	92829.730136948
Odisha	339928653.75	3680	92371.91677989133
Telangana	334332600.93	3636	91950.66032178218
Tamil Nadu	330103802.81	3602	91644.58712104386
Rajasthan	325159784.29	3555	91465.48081293952
West Bengal	315209940.81	3503	89982.85492720525



6. Statistical investigation

Mix effects: standardization across risk, loan type and DPD is close to actual March. Cohort effects are stratified by account age. Selection bias exists because targeting is observational. Survivorship is not safely measurable from the current account-status snapshot; status history should be used. Simpson checks compare within-stratum Feb vs Mar direction. Attribution-window checks show why last-touch channel attribution is not a safe causal metric. Time-series review shows March is not sustained through July.

dimension	common_strata	positive_mar_vs_feb	positive_share
risk_segment	4	2	0.5
loan_type	5	2	0.4
dpd_band	5	3	0.6
state	9	6	0.6666666666666666

7. Counterfactual

Campaign strategy versions overlap in time, so a historical pre/post targeting estimate is not identified cleanly. Recommended design: randomized account-level holdout or stepped-wedge rollout, stratified by DPD/risk/portfolio and optionally blocked by geography. Primary outcome: incremental 14- and 30-day recovery per eligible account; guardrails: complaints, contact burden and payer rate.

8. ₹10 Cr recommendation

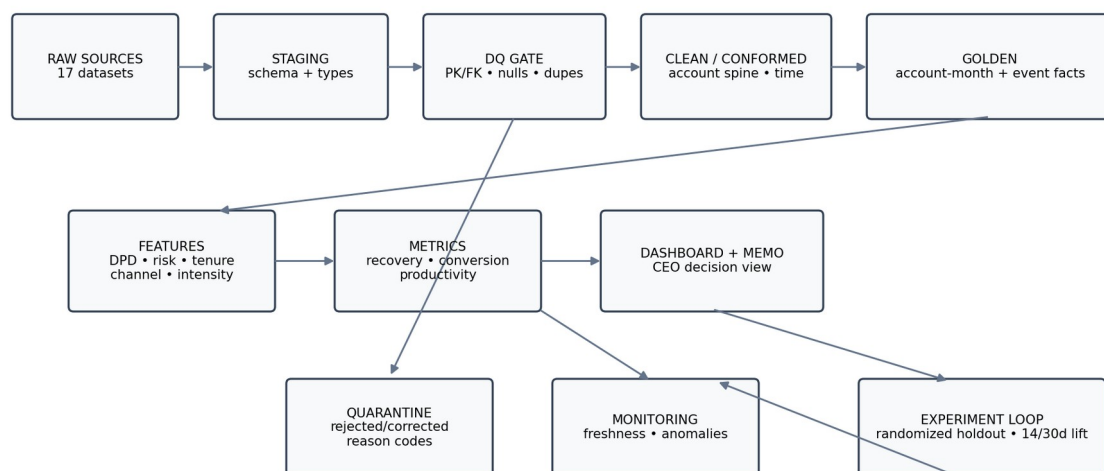
Recommendation: Better Borrower Targeting. Confidence is low for causal lift but high for the payment-ledger reconstruction. Release capital only through a controlled experiment with a pre-registered break-even gate.

lift	incremental_recovery_cr	gross_benefit_cost	net_roi_pct
0%	0.0	0.0	-100.0
2%	4.349054637702857	0.4349054637702857	-56.50945362297143
5%	10.872636594257145	1.0872636594257143	8.726365942571434
8%	17.39621855081143	1.7396218550811429	73.96218550811429

Annualized Jan-Jul clean recovery = ₹217.45 Cr; break-even lift = 4.60%. At 5% lift: ₹10.87 Cr incremental recovery, 1.09x gross benefit/cost, 8.7% net ROI. This is a planning scenario, not a causal forecast.

9. Production analytics design

COLLECTIONS ANALYTICS - PRODUCTION DESIGN



Data contracts • lineage • incremental loads • late arrivals • backfills • DQ alerts • anomaly detection

Controls: data contracts; explicit PK/FK rules; account_id event spine; effective-dated identity mappings; idempotent incremental processing; late-arrival watermarks; replayable backfills; quarantine reason codes; metric lineage; freshness/volume/null/duplicate checks; anomaly alerts.

10. Data gaps

dimension	available	source
Client	False	No client column in supplied schemas
Language	False	No language column in supplied schemas
Geography	True	borrowers.state, borrowers.city
Agent tenure	True	agents.joined_at + agent_sessions.login_at

Future data needed: client_id/client master, language, historical opening/closing balances, a full cost ledger, effective-dated agent/vendor identity history, and explicit treatment/control assignment.